PH 216, Problem set 1, Due 4/6

1. In class, we estimated the ground-state energy of the 3D SHO (see Shankar 12.6.42) using a 'guess' of

$$\psi_0(a;\vec{r}) = R(a;r)Y_0^0(\theta,\phi),$$

by varying a and minimizing $\langle\psi(a)|H|\psi(a)\rangle$.

Suppose we try to get the first energy E_1 using the trial function

$$\psi_1 = R(a;r)Y_1^0,$$

Where the Y enforces orthogonality with our ψ_0. A trial function with zero nodes that has the right asymptotics ($\propto r^{+1}$ as $r \to 0$ and $\to 0$ as $r \to \infty$) is

$$\psi_{2,1,0} = N(r/a_0)\exp(-r/2a_0)\cos\theta.$$

what then is E_1, and how does this compare with the first level of the actual 3D SHO?

Solution: We have

$$E(a) = \langle\psi(a)|H|\psi(a)\rangle = \langle\psi(a)|T|\psi(a)\rangle + \langle\psi(a)|V|\psi(a)\rangle.$$

As in class, we can use the quantum virial theorem,

$$\langle T\rangle = -\frac{1}{2}\langle V\rangle \to \langle T\rangle + \langle V\rangle = E_n = -\hbar^2/2ma^2n^2 = -\langle T\rangle$$

for an energy state of energy E_n. So we just have to evaluate

$$\langle V\rangle = \frac{1}{32\pi a^3}\int_0^\infty r^2\,dr \int d\Omega\,(\omega r^2/2)(r/a)^2\exp(-r/a)\cos^2\theta,$$

where I have taken the normalization of the wavefunction from Shankar 13.1.27. But since V does nothing to angles, the angular integral just falls out as 4π, so we get

$$\langle V\rangle = \frac{m\omega^2}{48\pi a^5}\int_0^\infty dr\,r^6\exp(-r/a) = \frac{m\omega^2}{48\pi a^5}\times 720a^6 = \frac{15a^2m\omega^2}{\pi}.$$

Combining our results leads to $E(a)$ written as:

$$E(a) = \frac{\hbar^2}{8ma^2} + 15a^2m\omega^2$$

With this in hand, we minimize the energy with respect to a:

$$\frac{dE}{da} = -\frac{\hbar^2}{4ma^3} + 30am\omega^2 = 0,$$

which gives

$$a_{min} = \left(\frac{\hbar^2}{120m^2\omega^2}\right)^{1/4}.$$

Thus, we get

$$E_1 = E(a_{min}) = \sqrt{\frac{15}{2}}\hbar\omega \approx 2.74\hbar\omega.$$

The exact result for the nth energy level of the 3D SHO is given by

$$E_n = (n + 3/2)\hbar\omega,$$

and with $n = 1$, $E_1 = 2.5\hbar\omega$. This gives $\Delta E_1 \approx 0.24\hbar\omega$ which is greater than zero as required by the variational principle.

2. (Shankar 16.2.8) Consider the $\ell = 0$ radial equation for the three-dimensional Coulomb problem. Since $V(r)$ is singular at the turning point $r = 0$, we cannot use $n + \frac{3}{4}$ as in eq. (16.2.42) of Shankar.

   (a) Will the additive constant be more or less than $\frac{3}{4}$? (Only heuristic reasoning is required, but you are welcome to derive the result formally if you like.)

   (b) By analyzing the exact equation near $r = 0$, it can be shown that the constant equals 1. Using this constant, show that the WKB energy levels agree with the exact results

Solution:

(a) Heuristically, as discussed in class if the wavefunction must vanish at both ends, the accumulated phase must add up to n half-cycles, so $(n + 1)\pi$. But if the wavefunction need not vanish, then you fit somewhat less, and the WKB formalism reveals that you basically take another $\pi/4$ cycles past the turning points, so $(n + 1/2)\pi$ or $(n + 3/4)\pi$ for zero or one endpoints fixed at $\psi = 0$.

For the H atom with $l = 0$, it's unclear to me intuitively what the condition should be. Near the origin we know that $\psi \propto r$, so it vanishes, and in fact even in an infinite square well in 1D, the wavefunction vanishes linearly near the endpoints. For $l = 1$, there is an infinite angular momentum barrier and the wavefunction vanishes as r^2 near $r = 0$. This might suggest $(n + 3/4)\pi$ for both cases, but it is not quite clear, since for $l = 0$ the equation is singular. To get the answer we can also analyze the radial equation more carefully, as follows.

We need to determine the asymptotic ($r \to \infty$) behavior of the $\ell = 0$ hydrogen atom wave function in the semi-classical limit. Since the semi-classical limit corresponds to large quantum numbers, $n \gg 1$ (where the bound state energy approaches zero), we can solve the Schrodinger equation for the hydrogen atom with $\ell = 0$ in the limit of $E \to 0$. Defining the reduced radial wave function, $u(r) \equiv rR(r)$, eq.(13.1.2) on p. 353 of Shankar (with $\ell = E = 0$) reads:

$$\left\{ \frac{d^2}{dr^2} + \frac{2me^2}{\hbar^2 r} \right\} u(r) = 0\,, \qquad (1)$$

with boundary conditions

$$u(r = 0) = 0\,, \qquad u(r \to \infty) = 0\,. \qquad (2)$$

It is convenient to define

$$\rho \equiv \frac{2me^2 r}{\hbar^2}\,.$$

Eq. (1) then reduces to

$$\frac{d^2u}{d\rho^2} + \frac{u(\rho)}{\rho} = 0\,. \tag{3}$$

This can be converted to a more recognizable form by defining

$$\rho = \tfrac{1}{4}z^2 \qquad \text{and} \qquad u(\rho) = zv(z)\,.$$

Then,

$$\frac{du}{d\rho} = \frac{du}{dz}\frac{dz}{d\rho} = \frac{2}{z}\frac{du}{dz}\,,$$

$$\frac{d^2u}{d\rho^2} = \frac{d}{d\rho}\left(\frac{2}{z}\frac{du}{dz}\right) = \frac{d}{dz}\left(\frac{2}{z}\frac{du}{dz}\right)\frac{dz}{d\rho} = \frac{4}{z^2}\left[\frac{d^2u}{dz^2} - \frac{1}{z}\frac{du}{dz}\right]\,.$$

Changing variables from ρ to z,

$$\frac{du}{dz} = \frac{d}{dz}(zv) = z\frac{dv}{dz} + v\,,$$

$$\frac{d^2u}{dz^2} = \frac{d}{dz}\left(z\frac{dv}{dz} + v\right) = z\frac{d^2v}{dz^2} + 2\frac{dv}{dz}\,.$$

It then follows that:

$$\frac{d^2u}{d\rho^2} = \frac{4}{z^2}\left[z\frac{d^2v}{dz^2} + \frac{dv}{dz} - \frac{v}{z}\right]\,.$$

Inserting this last result into eq. (3) yields:

$$z^2\frac{d^2v}{dz^2} + z\frac{dv}{dz} + \left(z^2 - 1\right)v(z) = 0\,.$$

We recognize this as Bessel's equation of order 1.[1] The general solution is:

$$v(z) = AJ_1(z) + BN_1(z)\,,$$

where A and B are coefficients to be determined. The boundary condition at $r = 0$ [cf. eq. (2)] implies that $B = 0$. The constant A is determined by the normalization condition. Hence, the reduced radial wave function is given by:

$$u(\rho) = 2A\sqrt{\rho}\,J_1(2\sqrt{\rho})\,.$$

The asymptotic form for the Bessel function $J_1(z)$ is given by:

$$J_1(z) \sim \sqrt{\frac{2}{\pi z}}\cos\left(z - \tfrac{3}{4}\pi\right)\,, \qquad \text{as } \; z \to \infty\,.$$

[1] More generally, Bessel's equation of order p reads: $z^2v'' + zv' + (z^2 - p^2)v = 0$, with general solution $v(z) = AJ_p(z) + BN_p(z)$.

Hence,

$$u(\rho) \sim 2A\sqrt{\rho}\,\cos\left(2\sqrt{\rho} - \tfrac{3}{4}\pi\right)\,, \qquad \text{as} \;\; \rho \to \infty\,. \tag{4}$$

To match this to a WKB wave function, we note that

$$k(r) = \frac{1}{\hbar}\left[2m\left(E + \frac{e^2}{r}\right)\right]^{1/2}\,. \tag{5}$$

In the semi-classical limit where $E \simeq 0$, we simply obtain

$$k(r) \simeq \sqrt{\frac{2me^2}{\hbar^2 r}} = \frac{2me^2}{\hbar^2}\rho^{-1/2}\,,$$

and

$$\int_0^r k(r')dr' \simeq \int_0^\rho \frac{d\rho'}{\sqrt{\rho'}} = 2\rho^{1/2}\,.$$

Hence, one can rewrite eq. (4) as:

$$u(r) \sim \frac{2A}{[k(r)]^{1/2}}\cos\left(\int_0^r k(r')dr' - \frac{3\pi}{4}\right)\,, \qquad \text{as} \;\; r \to \infty\,.$$

In class, we argued that for a spherically symmetric potential that is regular at the origin, the asymptotic form of the reduced radial wave function in the region $0 < r < r_c$, where r_c is the turning point closest to the origin, is given by

$$u(r) \sim \frac{C}{[k(r)]^{1/2}}\cos\left(\int_0^r k(r')dr' - \frac{\pi}{2}\right)\,, \qquad \text{as} \;\; r \to \infty\,.$$

Thus, the phase shift is *increased* by $\frac{1}{4}\pi$. This means that in the derivation of the energy quantization condition [eq. (16.2.42) of Shankar], the factor of $(n + \frac{3}{4})\pi$ must be increased by $\frac{1}{4}\pi$. The correct quantization condition for the $\ell = 0$ bound states of the Coulomb potential is thus given by:

$$\boxed{\int_0^{r_c} k(r)dr = (n+1)\pi\,, \qquad n = 0, 1, 2, 3, \ldots} \tag{6}$$

(b) We now use eq. (6) to compute the energies of the $\ell = 0$ bound states of the Coulomb potential. The turning point r_c is obtained by setting $k(r_c) = 0$, where $k(r)$ is given by eq. (5). Thus, $E + e^2/r_c = 0$, which yields

$$r_c = -\frac{e^2}{E}\,.$$

Using eq. (6),

$$\frac{1}{\hbar}\int_0^{-e^2/E}\left[2m\left(E + \frac{e^2}{r}\right)\right]^{1/2} dr = (n+1)\pi\,. \tag{7}$$

Keep in mind that $E < 0$ for the bound state energy levels. Defining the dimensionless real positive variable,

$$x \equiv -\frac{rE}{e^2}\,,$$

eq. (7) can be rewritten as

$$\left(-\frac{2me^4}{\hbar^2 E}\right)^{1/2} \int_0^1 \left(\frac{1}{x} - 1\right)^{1/2} dx = (n+1)\pi\,, \tag{8}$$

The integral is recognized as a beta function:

$$\int_0^1 \left(\frac{1}{x} - 1\right)^{1/2} dx = \int_0^1 x^{-1/2}(1-x)^{1/2}\, dx = B\left(\tfrac{1}{2}, \tfrac{3}{2}\right) = \frac{\Gamma(\frac{1}{2})\Gamma(\frac{3}{2})}{\Gamma(2)} = \tfrac{1}{2}\pi\,,$$

where we have used $\Gamma(\frac{3}{2}) = \frac{1}{2}\Gamma(\frac{1}{2})$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Inserting the above result into eq. (8) and solving for E, we obtain

$$\boxed{E = \frac{-me^4}{2\hbar^2(n+1)^2}\,, \qquad n = 0, 1, 2, 3, \ldots}$$

which coincide with the exact energy levels of the hydrogen atom!

3. The interactions of heavy quarks are often approximated by a spin-independent non-relativistic potential that is a linear function r, the separation between the quarks: $V(r) = A + Br$. Thus the "Charmonium" particles, the ψ and ψ', with rest energies of 3.1 GeV and 3.7 GeV, are believed to be the $n = 0$ and $n = 1$ zero-angular momentum bound states of a charm quark and antiquark of mass $m_c c^2 = 1.5\, GeV$. Similarly, the particles Υ (energy 9.5 GeV) and Υ' are believed to be the $n = 0$ and $n = 1$ zero-angular-momentum bound states of the 'bottom' quark and antiquark, which have rest mass $m_b c^2 = 4.5\, GeV$.

   (a) Using dimensional analysis, derive a relationship between the energy splitting between, on the one hand, ψ and ψ' and, on the other hand, Υ and Υ'. Use this to find the energy of the Υ'.

   (b) Call the $n = 2$, zero angular-momentum charmonium particle the ψ''. Use the WKB approximation to estimate the energy splitting of the ψ' and ψ'' in terms of the energy splitting between the ψ and the ψ', and then use this to give a numerical estimate of the rest energy of the ψ''.

Solution:

In the center-of-mass system of a quark and it antiquark, the equation of relative motion is

$$\left[-\frac{\hbar^2}{2\mu}\nabla_r^2 + V(r)\right]\psi(\mathbf{r}) = E_R\psi(\mathbf{r}), \quad \mu = m_q/2,$$

where E_R is the relative motion energy, m_q is the mass of the quark. When the angular momentum is zero, the above equation in spherical coordinates can be simplified to

$$\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right) + V(r)\right]R(r) = E_R R(r).$$

Let $R(r) = \chi_0(r)/r$. Then $\chi_0(r)$ satisfies

$$\frac{d^2\chi_0}{dr^2} + \frac{2\mu}{\hbar^2}[E_R - V(r)]\chi_0 = 0,$$

i.e.,

$$\frac{d^2\chi_0}{dr^2} + \frac{2\mu}{\hbar^2}(E_R - A - Br)\chi_0 = 0.$$

(a) Suppose the energy of a bound state depends on the principal quantum number n, which is a dimensionless quantity, the constant B in $V(r)$, the quark reduced mass μ, and $\hbar$, namely

$$E = f(n)B^x\mu^y\hbar^x.$$

As,

$$[E] = [M][L]^2[T]^{-2},$$
$$[B] = [M][L][T]^{-2}, \quad [\mu] = [M]$$
$$[\hbar] = [M][L]^2[T]^{-1},$$

we have

$$x = z = \frac{2}{3}, y = -\frac{1}{3}$$

and hence

$$E = f(n)(B\hbar)^{2/3}(\mu)^{-1/3},$$

where $f(n)$ is a function of the principal quantum number n. Then

$$\Delta E_\psi = E_{\psi'} - E_\psi = f(1)\frac{(B\hbar)^{2/3}}{\mu_c^{1/3}} - f(0)\frac{(B\hbar)^{2/3}}{\mu_c^{1/3}}$$

$$= \frac{(B\hbar)^{2/3}}{\mu_c^{1/3}}[f(1) - f(0)],$$

and similarly

$$\Delta E_\Upsilon = \frac{(B\hbar)^{2/3}}{\mu_b^{1/3}}[f(1) - f(0)].$$

Hence

$$\frac{\Delta E_\Upsilon}{\Delta E_\psi} = \left(\frac{\mu_c}{\mu_b}\right)^{1/3} = \left(\frac{1}{3}\right)^{1/3}.$$

As

$$E_{\Upsilon'} - E_\Upsilon \approx 0.42 \text{ GeV},$$

$$E_{\Upsilon'} = E_\Upsilon + 0.42 \text{ GeV} = 9.5 \text{ GeV} + 0.42 \text{ GeV} \approx 9.9 \text{ GeV}.$$

(b) Applying the WKB Approximation to the equation for χ_0 we obtain the Bohr-Sommerfeld quantization rule

$$2\int_0^\alpha \sqrt{2\mu(E_r - A - Br)}dr = (n + 3/4)\hbar \quad \text{with} \quad \alpha = \frac{E_R - A}{B}$$

which gives, writing E_n for E_R,

$$E_n = A + \frac{[3(n+\frac{3}{4})B\hbar/4]^{2/3}}{(2\mu)^{1/3}}.$$

Application to the energy splitting gives

$$E_{\psi'} - E_{\psi} = \frac{(B\hbar)^{2/3}}{\mu_c^{1/3}}\left[\left(\frac{21}{16}\right)^{2/3} - \left(\frac{9}{16}\right)^{2/3}\right],$$

$$E_{\psi''} - E_{\psi'} = \frac{(B\hbar)^{2/3}}{\mu_c^{1/3}}\left[\left(\frac{33}{16}\right)^{2/3} - \left(\frac{21}{16}\right)^{2/3}\right],$$

and hence

$$\frac{E_{\psi''} - E_{\psi'}}{E_{\psi'} - E_{\psi}} = \frac{(33)^{2/3} - (21)^{2/3}}{(21)^{2/3} - (9)^{2/3}} \approx 0.81.$$

Thus

$$E_{\psi''} - E_{\psi'} = 0.81 \times (E_{\psi'} - E_{\psi}) = 0.81 \times (3.7 - 3.1)\ \text{GeV} \approx 0.49\ \text{GeV},$$

and

$$E_{\psi''} = 3.7\ \text{GeV} + 0.49\ \text{GeV} \approx 4.2\ \text{GeV}.$$